WEEK 7 OVERVIEW

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CEVE 543

2026-10-05



TRENDS AT NEIGHBORING TEXAS STATIONS

Note

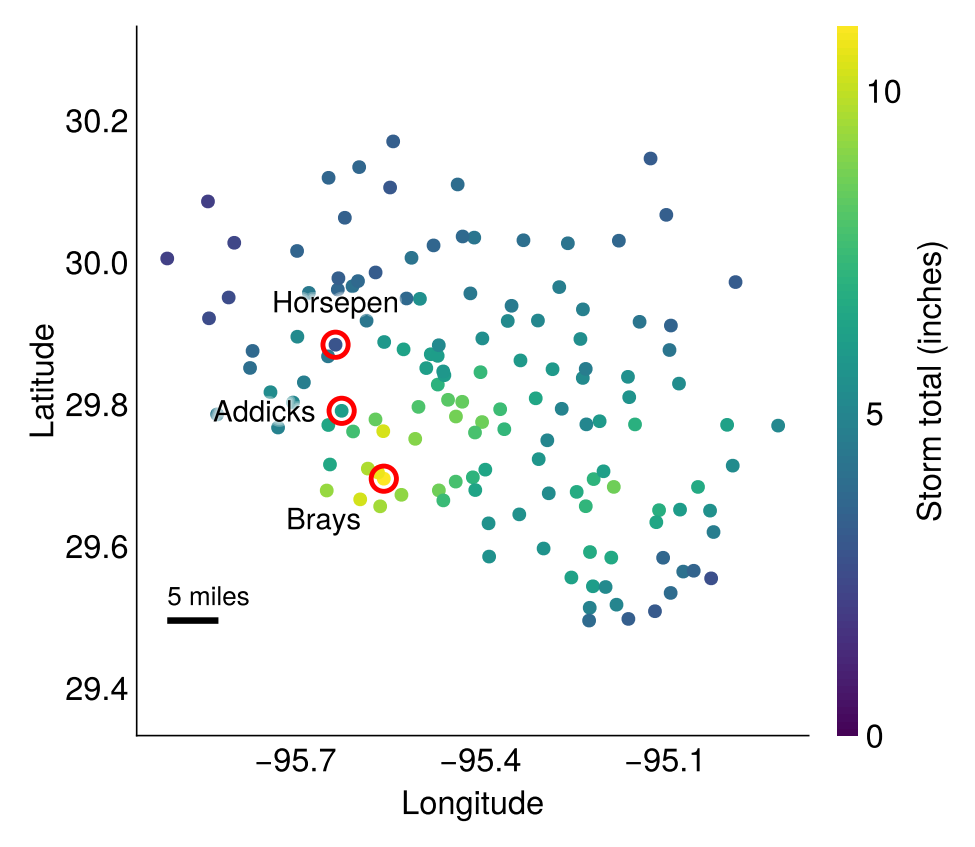
Figure shown in class: Kendall's rank correlation between global temperature and 1-hour (left) and 1-day (right) annual maximum rainfall at Texas stations; blue is positive and red negative (preliminary, from a paper in preparation in my group).

CO2 TRENDS FIT STATION BY STATION

Note

Figure shown in class: The coefficient on CO2 in the location (top) and scale (bottom) of a nonstationary extreme value distribution, fit separately at each of 181 Gulf Coast stations; purple is an increase with CO2 and orange a decrease (preliminary, from a paper in preparation in my group).

MEMORIAL DAY 2015

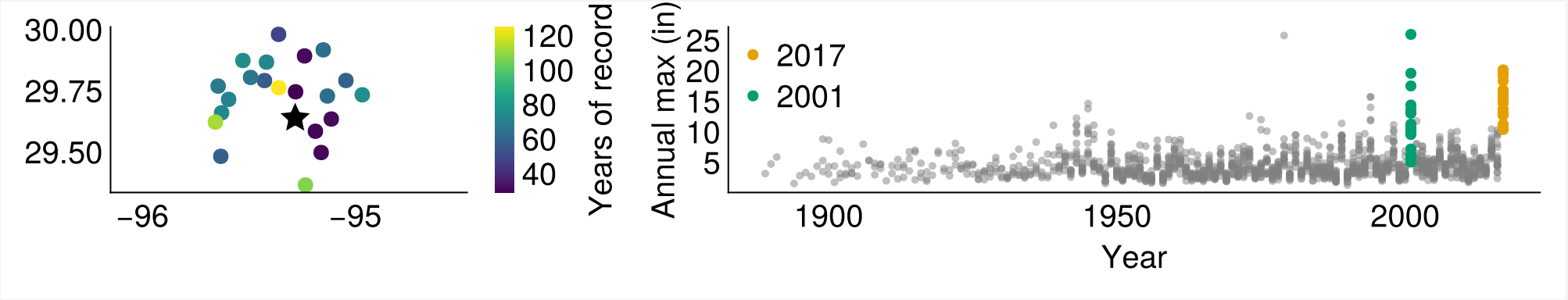


Totals for May 25-26 at 138 Harris County Flood Control District gauges.

- Brays Bayou: 11.00 inches.
- Addicks Dam, 7.8 miles from Brays: 6.08 inches.
- Horsepen Creek, 13.9 miles from Brays: 2.84 inches.

Within 14 miles, one storm's totals ranged from 2.84 to 11.00 inches.

ANNUAL MAXIMUM RAINFALL NEAR HOBBY AIRPORT



- 22 Atlas 14 stations within 25 miles of Hobby Airport (star) (Perica et al., 2018).
- 16 had their largest year in 2017 (Harvey), and 4 in 2001 (Allison).
- Fit alone, 100-year estimates run from 12.2 to 37.4 inches; the five largest come from 29- to 31-year records.

OBJECTIVES

If nearby stations see the same storms, pooling their records should average out where each storm landed.

By the end of this week, you can:

1. Describe the sources of noise in one station's record of extremes.
2. Describe, physically and statistically, the bias-variance trade-off in regionalization.
3. State at least three ways to regionalize and the key assumptions each makes.
4. Read a regional extremes analysis and assess whether its key assumptions hold.